COMPANION PRACTICE BOOK

Fresh questions. Fully worked.

Jane Street TDOE preparation

15 new problems across the five practice areas you liked. Every question is followed immediately by its answer and a complete explanation.

BUILT FOR UNDERSTANDING

Each two-page spread starts with the question, the direct answer and the main solution. The next page works through the follow-ups, shows how to explain the reasoning aloud, and checks the result.

The approach on every problem

Answer See the destination before working through the method.

Explanation Understand what to model, why the calculation works, and how each step follows.

Interview reasoning Practise explaining the model, calculation and checks in your own words.

All problems in this companion are authored practice. They are new to this conversation and are not represented as verified Jane Street interview questions. Topic selection builds on the research in the first pack. The previous pack's Set 3 is omitted.

NAVIGATION

Choose a question

Click a question title to jump to its question-and-answer page. Each spread is self-contained.

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A PRACTICAL WAY TO USE THE BOOK

Try the question aloud before reading down. If you need help, read one numbered step, pause, and attempt the next. After finishing, cover the explanation and reconstruct the logic. The spoken walkthrough is an example of clear reasoning, not a script to memorise.

The five sets cover invariants, counting and conditional probability, expected value, rates, and number structure. The final source note explains how this practice relates to the research.

QUESTION 01 / A / INVARIANTS AND STRUCTURED COUNTING

Mixing two salt solutions

Conservation, weighted averages and changing percentages | Authored practice

The question

You have a solution that is 20% salt by mass and another that is 50% salt by mass. How many grams of each should you mix to make 600 g of a 30% salt solution? Assume no material is lost and the mixture becomes uniform.

THE ANSWER

Mix 400 g of the 20% solution with 200 g of the 50% solution.

How to work it out

1. Start with the quantity that must balance
The final mixture must contain 30% of 600 g as salt: 0.30 x 600 = 180 g. Mixing changes the concentration, but it neither creates nor destroys salt. So the salt contributed by the two inputs must add to 180 g. This is the equation to build; averaging the two printed percentages would ignore how much of each solution we use.

2. Use one unknown, because the total is fixed
Let x be the grams of the stronger, 50% solution. The weaker solution must then weigh 600-x grams. The strong portion contributes 0.50x grams of salt. The weak portion contributes 0.20(600-x) grams. Both expressions now measure the same thing: grams of salt.

3. Write and solve the balance
Set 0.50x + 0.20(600-x) = 180. Expanding gives 0.50x + 120 - 0.20x = 180. Collecting the x terms gives 0.30x = 60, so x = 200. The remaining mass is 600-200 = 400 g of the weaker solution.

4. See an intuitive route to the same answer
Imagine first making all 600 g from the 20% solution. That would contain 120 g of salt, which is 60 g short. Replacing one gram of that solution with one gram of the 50% solution adds 0.50-0.20 = 0.30 g of salt. To add the missing 60 g, replace 60/0.30 = 200 g. This gives the same mixture without starting from an algebraic formula.

QUESTION 01 / CONTINUED

Follow-ups and interview reasoning

Mixing two salt solutions | Back to the question

Follow-up 1. Remove 150 g of the mixture and replace it with 150 g of pure water. What is the new salt percentage?

Because the mixture is uniform, the removed portion is also 30% salt. Removing 150 g therefore removes 0.30 x 150 = 45 g of salt. We started with 180 g, so 135 g remains. Adding pure water restores the total mass to 600 g but adds no salt. The new fraction is 135/600 = 0.225, or 22.5%. The key is to track salt mass and total mass separately.

Follow-up 2. How much 50% solution must now be added to restore a 30% concentration?

Let y grams be added. Salt mass becomes 135+0.50y, while total mass becomes 600+y. Therefore (135+0.50y)/(600+y) = 0.30. Multiply through: 135+0.50y = 180+0.30y. Thus 0.20y = 45 and y = 225 g. Check: the final mixture has 247.5 g of salt in 825 g, and 247.5/825 = 0.30. The total is allowed to grow in this follow-up; we are adding rather than replacing.

HOW YOU COULD EXPLAIN IT ALOUD

I would balance grams of salt. The target contains 180 g. If x grams come from the stronger solution, the rest is 600-x, so the salt equation is 0.5x + 0.2(600-x) = 180. That gives 200 g of strong solution and 400 g of weak. I would check that the two salt contributions, 100 g and 80 g, add to 180 g. For the replacement follow-up, I would update the salt mass first and only then calculate the new percentage.

Check the result. The final 30% lies between 20% and 50%, and the mixture contains more of the weaker solution. That makes sense because 30% is closer to 20%.

Avoid this mistake. A simple average, (20%+50%)/2 = 35%, assumes equal masses. In the last follow-up, keeping 600 as the denominator would forget the mass of the added solution.

Use this idea elsewhere. When percentages change, ask what is being conserved and write every term in actual units before taking a ratio.

QUESTION 02 / A / INVARIANTS AND STRUCTURED COUNTING

A tournament with an awkward number of players

Counting eliminations instead of simulating rounds | Authored practice

The question

A single-elimination tournament has 73 players. Every match has exactly one winner and one loser; a loser is eliminated, and the winner continues. Byes are allowed. How many matches are needed to produce one champion?

THE ANSWER

Exactly 72 matches, regardless of how the byes are arranged.

How to work it out

1. Identify what one match changes
At the start, 73 players remain eligible to become champion. Every completed match removes exactly one player from that group: the loser. The winner remains eligible, so a match reduces the number of remaining players by one, not two.

2. Work backwards from the final state
At the end, exactly one player remains. The number of players that must be eliminated is therefore 73-1 = 72. Since each match causes one elimination, there must be exactly 72 matches. This counts the required change directly and avoids choosing a particular bracket.

3. Explain why byes do not alter the count
A bye advances a player without eliminating anyone. It changes the scheduling of the tournament, but does not contribute to the required 72 eliminations. However the bracket is arranged, every non-champion must eventually lose one match. Counting those losing players gives the same answer.

4. Check a tiny example
With three players, one player can receive a bye while the other two play. Their winner then faces the player with the bye. There are two matches, which is 3-1. With one player, there are zero matches, also consistent with the formula. These cases test the reasoning at both a small non-power-of-two size and its boundary.

QUESTION 02 / CONTINUED

Follow-ups and interview reasoning

A tournament with an awkward number of players | Back to the question

Follow-up 1. Arrange an opening round that leaves exactly 64 players. How many opening matches and byes are needed?

We must reduce the field from 73 to 64, which requires nine eliminations and therefore nine matches. Those matches use 18 players. The other 73-18 = 55 players receive byes. The next field contains nine winners plus 55 bye recipients, totalling 64. From there, a complete 64-player bracket needs 63 more matches, so the overall total is 9+63 = 72.

Follow-up 2. What is the general result for n players?

There are n-1 matches, provided every match eliminates exactly one player and the event ends with one champion. This is a statement about those rules. If players can re-enter, a match can have no loser, or a second loss is required for elimination, the same argument cannot be applied unchanged.

HOW YOU COULD EXPLAIN IT ALOUD

I do not think I need to build the bracket. Every match eliminates one player, and 72 of the 73 players must be eliminated. So the answer is 72. Byes do not eliminate anyone, so they do not affect that total. If you want an explicit construction, nine opening matches and 55 byes leave 64 players; the remaining bracket then needs 63 matches. That also totals 72.

Check the result. Count losers, not player appearances. There is one losing appearance per match and each non-champion has exactly one losing appearance.

Avoid this mistake. Rounding 73 up to 128 and reporting 127 matches counts matches involving imaginary entrants. A bye is not a played match.

Use this idea elsewhere. When an event happens repeatedly, ask whether each occurrence changes a simple count by a fixed amount. The total change may solve the problem immediately.

QUESTION 03 / A / INVARIANTS AND STRUCTURED COUNTING

Routes through a grid with a blocked junction

Ordered choices, combinations and counting the complement | Authored practice

The question

On a grid, you travel from (0,0) to (4,3). Each move is one unit right or one unit up. The junction (2,1) is blocked. How many different routes avoid that junction?

THE ANSWER

17 routes avoid the blocked junction. There are 35 routes in total and 18 that pass through it.

How to work it out

1. Turn a route into a sequence
Every route contains exactly four right moves and three up moves, in some order. For example, RRUURRU is one valid seven-move route. Once the seven-letter sequence is specified, the route is specified. We can therefore count sequences rather than draw every path.

2. Count all routes before imposing the restriction
Choose which three of the seven positions contain U; the other four must contain R. This gives C(7,3) = (7 x 6 x 5)/(3 x 2 x 1) = 35. We divide by 3! because choosing the same three positions in a different order does not create a new route.

3. Count the routes that hit the blocked junction
Any route through (2,1) splits uniquely into a route to that point and a route from it to (4,3). The first part requires two right moves and one up move, so there are C(3,1) = 3 choices. The second requires two right moves and two up moves, so there are C(4,2) = 6 choices.

4. Multiply the two stages, then subtract
Each of the three possible first parts can be joined to each of the six second parts. Thus 3 x 6 = 18 routes pass through the junction. No route is counted twice, because its prefix and suffix at that fixed point are uniquely determined. Subtract from the unrestricted total: 35-18 = 17 routes avoid it.

QUESTION 03 / CONTINUED

Follow-ups and interview reasoning

Routes through a grid with a blocked junction | Back to the question

Follow-up 1. Choose uniformly from all 35 unrestricted routes. What is the probability the chosen route avoids the junction?

Exactly 17 of the 35 equally likely routes avoid it, so the probability is 17/35. Notice the sampling rule: choosing uniformly among complete routes is specified. Choosing right or up with equal probability at every available junction is a different random procedure and need not make all complete routes equally likely.

Follow-up 2. Move the destination to (5,3), leaving the blocked junction at (2,1). What changes?

The unrestricted route now contains five rights and three ups, giving C(8,3) = 56 routes. The prefix to the blocked junction still has three choices. From (2,1) to (5,3), there are three rights and two ups, giving C(5,2) = 10 suffixes. Therefore 3 x 10 = 30 routes hit the junction and 56-30 = 26 avoid it. Only the part of the calculation after the changed destination needed updating.

HOW YOU COULD EXPLAIN IT ALOUD

I will first count all routes and subtract the ones that use the blocked point. Each full route is an ordering of four R moves and three U moves, so there are 35. A route through (2,1) has three possible prefixes and six possible suffixes. That gives 18 blocked routes, leaving 17. The multiplication is valid because every prefix can be joined to every suffix, and the split at the blocked point is unique.

Check the result. The answer must be between zero and 35. If the blocked point were the start or destination, every route would be excluded. If it were outside the reachable rectangle, none would be excluded.

Avoid this mistake. Using 7! treats the four identical right moves and three identical up moves as distinguishable. Counting routes through the point requires multiplying the two stages, not adding them.

Use this idea elsewhere. For a hard restriction, first ask whether the unrestricted set and the forbidden set are easier to count separately.

QUESTION 04 / B / COUNTING AND CONDITIONAL PROBABILITY

A PIN with exactly one repeated pair

Avoiding double counting and defining the sample space | Authored practice

The question

A four-digit PIN is generated by choosing each digit independently and uniformly from 0 to 9. Leading zeros are allowed. What is the probability that exactly one digit appears twice and the other two digits appear once each and differ from one another? For example, 0073 qualifies; 0007 and 0077 do not.

THE ANSWER

There are 4,320 qualifying PINs out of 10,000, so the probability is 0.432, or 43.2%.

How to work it out

1. Pin down the required pattern
The multiplicities must be 2,1,1. One value is repeated, while the two singleton values are different from that repeated value and from each other. This excludes three-of-a-kind, four-of-a-kind and two separate pairs. Giving qualifying and non-qualifying examples helps ensure the interpretation is settled.

2. Choose the repeated value and its locations
There are ten choices for the digit that is repeated. Choose two of the four PIN positions for that digit: C(4,2) = 6 choices. Once the repeated digit and those two positions are fixed, there are two empty positions left.

3. Fill the remaining positions in a fixed order
Take the leftmost empty position first. It can contain any of the nine digits other than the repeated value. The remaining empty position then has eight choices, because it must differ from both values already used. Hence the count is 10 x 6 x 9 x 8 = 4,320. There is no extra factor of two: the 9 x 8 choices already distinguish the two singleton positions.

4. Divide by all equally likely PINs
There are 10 choices independently for each of four positions, so 10^4 = 10,000 total PINs. The probability is 4,320/10,000 = 0.432. Every qualifying PIN is counted once, because it has one identifiable repeated value, one identifiable pair of repeated positions and one fixed pair of singleton values in their positions.

QUESTION 04 / CONTINUED

Follow-ups and interview reasoning

A PIN with exactly one repeated pair | Back to the question

Follow-up 1. What if a four-digit integer is selected uniformly from 1000 to 9999 instead?

Now the first digit cannot be zero. Among qualifying PINs, each of the ten digits is equally likely to occupy the first position: relabelling the digits does not change the 2,1,1 pattern. Therefore exactly one tenth of the 4,320 qualifying PINs start with zero. Excluding those leaves 4,320 x 9/10 = 3,888. The total number of four-digit integers is 9,000. The probability is 3,888/9,000 = 0.432 again. Both numerator and denominator shrink by the same proportion.

Follow-up 2. Can you count the same PINs another way?

Choose the repeated digit in ten ways, then choose the two different singleton values as an unordered pair from the remaining nine digits: C(9,2) = 36. Arrange the four values in 4!/2! = 12 distinct ways, dividing by 2! for the identical repeated copies. The result is 10 x 36 x 12 = 4,320. This agrees with the position-first method.

HOW YOU COULD EXPLAIN IT ALOUD

I will interpret exactly one pair as a 2,1,1 pattern. I can choose the paired digit in ten ways and its two positions in six ways. The empty positions then have nine and eight choices when filled from left to right. That gives 4,320 PINs out of 10,000. I would check that I have not multiplied by two again for swapping the singleton values, because their order was already counted in the nine-times-eight step.

Check the result. For the integer version, verify 3,888/9,000 = 4,320/10,000. The equality follows from digit-label symmetry, not from a rule that leading-zero restrictions never matter.

Avoid this mistake. The common error is to count ordered singleton values and then also multiply by their two possible orders. Choose an ordered method or an unordered method and stay consistent.

Use this idea elsewhere. A clean counting construction should make it possible to reconstruct exactly one set of choices from any final object.

QUESTION 05 / B / COUNTING AND CONDITIONAL PROBABILITY

Which bag did the red ball come from?

Updating probabilities and predicting the next observation | Authored practice

The question

Bag A contains two red balls and one blue ball. Bag B contains one red ball and three blue balls. You choose a bag with probability 1/2 each, then draw one ball uniformly from that bag. The ball is red. What is the probability you chose Bag A?

THE ANSWER

The probability the bag is A, after seeing red, is 8/11.

How to work it out

1. Separate the choice of bag from the colour
Before the draw, A and B are equally likely. But red is more likely from A: its red fraction is 2/3, compared with 1/4 in B. Seeing red therefore gives evidence in favour of A. We should expect the updated probability of A to exceed 1/2.

2. Count an imagined batch of experiments
Imagine 24 experiments, with 12 choosing each bag. In the A group, the expected number of red draws is 12 x 2/3 = 8. In the B group, it is 12 x 1/4 = 3. Among the eleven red draws in this proportional picture, eight came from A. This gives 8/11. These are probability weights, not a claim that a real batch of 24 will produce exactly those counts.

3. Write the same reasoning as probabilities
The joint probability of choosing A and seeing red is (1/2)(2/3) = 1/3 = 8/24. The probability of choosing B and seeing red is (1/2)(1/4) = 1/8 = 3/24. Total red probability is 11/24. Restricting to red outcomes gives P(A | red) = (8/24)/(11/24) = 8/11.

4. Keep the source of randomness clear
It would be wrong to pool the balls and use two of the three red balls. The experiment chooses bags equally, even though the bags contain different numbers of balls. An individual ball in A has probability (1/2)(1/3) = 1/6 of being drawn; an individual ball in B has probability (1/2)(1/4) = 1/8. Individual balls across the two bags are not equally likely.

QUESTION 05 / CONTINUED

Follow-ups and interview reasoning

Which bag did the red ball come from? | Back to the question

Follow-up 1. Without replacing the red ball, draw again from the same chosen bag. What is the chance the next ball is red?

After the first red, the bag probabilities are A: 8/11 and B: 3/11. If the bag is A, one red and one blue remain, so the next red probability is 1/2. If it is B, its only red ball has gone, so the next red probability is zero. Average over the two possible bags: (8/11)(1/2) + (3/11)(0) = 4/11.

Follow-up 2. What if the first red ball is replaced before the second draw?

The evidence about which bag was selected is unchanged: A is still 8/11 likely. But the bag contents are restored. The next red probability is now (8/11)(2/3) + (3/11)(1/4) = 16/33 + 3/44 = 64/132 + 9/132 = 73/132. Replacement changes the within-bag probabilities; it does not erase the information learned about the bag.

HOW YOU COULD EXPLAIN IT ALOUD

Red is more likely from A, so I expect the answer to be above one half. I would weight the two ways to see red: one half times two thirds from A, and one half times one quarter from B. Those weights are eight twenty-fourths and three twenty-fourths. Of the eleven total red weights, eight come from A, so the answer is eight elevenths. For another draw I would keep these updated bag probabilities and then update the remaining contents separately.

Check the result. 8/11 is between 1/2 and 1, as expected. Without replacement the next-red chance is 4/11, lower than with replacement, because a red ball has been removed.

Avoid this mistake. Do not reuse the original 1/2 bag probabilities after observing red. Also do not assume replacing the ball resets your knowledge of the chosen bag.

Use this idea elsewhere. For a hidden source, update how likely each source is, then predict future observations by averaging within-source probabilities using those updated weights.

QUESTION 06 / B / COUNTING AND CONDITIONAL PROBABILITY

Heads that must stay apart

Counting arrangements with adjacency restrictions | Authored practice

The question

Flip a fair coin six independent times. What is the probability of getting exactly three heads with no two heads adjacent? A head in the first position may not be next to a head in the second, and so on; the first and last positions are not neighbours.

THE ANSWER

There are four qualifying sequences, so the probability is 4/64 = 1/16.

How to work it out

1. Keep the two requirements separate
We need exactly three heads, so the other three flips must be tails. We also need the heads separated. The ordinary count C(6,3) = 20 includes arrangements such as HHHTTT that violate the separation condition, so it is not the favourable count here.

2. Place the tails first
Write the three tails as T T T. They create four gaps: before the first tail, between the first and second, between the second and third, and after the last tail. To avoid adjacent heads, put at most one head into each gap. If two heads were put in the same gap, nothing would separate them.

3. Choose three of the four gaps
We need to insert three heads, so choose three of the four gaps. There are C(4,3) = 4 choices. The resulting sequences are HTHTHT, HTHTTH, HTTHTH and THTHTH. Listing them once is a useful check that the gap method has neither missed a route nor counted the same sequence twice.

4. Use the probability of a complete sequence
Each specific sequence of six fair independent flips has probability (1/2)^6 = 1/64. The four favourable sequences are disjoint, so their probabilities add to 4/64 = 1/16. The events "three heads" and "no adjacent heads" should not be multiplied as if independent.

QUESTION 06 / CONTINUED

Follow-ups and interview reasoning

Heads that must stay apart | Back to the question

Follow-up 1. You are already told there were exactly three heads. What is the probability they were non-adjacent?

Conditioning changes the denominator. There are C(6,3) = 20 equally likely sequences with exactly three heads, and the same four qualify. The conditional probability is 4/20 = 1/5. The numerator did not change; the sample space shrank from all 64 sequences to the 20 consistent with the information.

Follow-up 2. Generalise to n fair flips and exactly k non-adjacent heads.

There are n-k tails. Placing them creates n-k+1 gaps. Choose k different gaps for the heads, giving C(n-k+1,k) qualifying sequences when there are at least k gaps. Their unconditional probability is C(n-k+1,k)/2^n. If k > n-k+1, there are too many heads to keep apart, so the count is zero. This method produces the formula directly; it does not require remembering it.

HOW YOU COULD EXPLAIN IT ALOUD

Since there must be three tails, I would place those first. They create four gaps, and each gap can hold at most one head if heads are to stay separated. Choosing three of those gaps gives four sequences. Each has probability one sixty-fourth, so the answer is one sixteenth. If I am told there were exactly three heads, I keep the same four favourable sequences but divide by the twenty sequences with three heads, giving one fifth.

Check the result. Three heads need at least two separating tails, so they are possible in six flips. Four non-adjacent heads in six flips would need three tails and therefore at least seven flips; the general formula correctly gives no valid arrangement.

Avoid this mistake. Do not require a tail at both ends: a head may occupy either end gap. Also do not treat the sequence as circular unless the question says so.

Use this idea elsewhere. For "no two adjacent" constraints, place the separating objects first and choose which gaps receive the constrained objects.

QUESTION 07 / C / EXPECTED VALUE AND DECISIONS

Which check should run first?

Expected cost, stopping early and comparing strategies | Authored practice

The question

An item must pass two checks. Check A takes 2 minutes and rejects 10% of items. Check B takes 5 minutes and rejects 50%. The two results are independent. You stop as soon as a check rejects the item; otherwise you perform both. Which order minimises the expected time spent checking one item?

THE ANSWER

Run B first. B then A takes 6 minutes on average; A then B takes 6.5 minutes.

How to work it out

1. Identify what is paid for certain
The first check always runs, so its entire time cost is unavoidable in that order. The second check runs only when the item passes the first. The average total is therefore the first check time plus the second check time multiplied by the probability of reaching it.

2. Calculate A followed by B
A always takes 2 minutes. An item passes A with probability 0.90, and only then do we spend 5 minutes on B. Expected time is 2 + 0.90 x 5 = 6.5 minutes. Equivalently, 10% of items take 2 minutes and 90% take 7 minutes: 0.10 x 2 + 0.90 x 7 = 6.5.

3. Calculate B followed by A
B always takes 5 minutes. Half the items are rejected immediately; the other half require the additional 2-minute A check. Expected time is 5 + 0.50 x 2 = 6 minutes. Equivalently, 0.50 x 5 + 0.50 x 7 = 6. The longer check is worth doing first here because it stops enough items from needing the other check.

4. Describe the tradeoff precisely
Running the faster test first saves time when it rejects an item, but A rejects only one item in ten. B rejects one in two. We should compare expected total time, not just which test is faster or which rejects more. Across 100 representative items, the expected saving is 100 x 0.5 = 50 minutes by putting B first.

QUESTION 07 / CONTINUED

Follow-ups and interview reasoning

Which check should run first? | Back to the question

Follow-up 1. What fraction of items pass both checks, and does the order change it?

Independence gives P(pass both) = 0.90 x 0.50 = 0.45, or 45%. Changing the order does not change which items satisfy both requirements. It changes only how much checking time is spent before discovering a rejection.

Follow-up 2. Find a general comparison rule.

Let cA and cB be check times and pA and pB be rejection probabilities. A first costs cA+(1-pA)cB on average; B first costs cB+(1-pB)cA. A first is better when cA+(1-pA)cB <= cB+(1-pB)cA. Cancel cA+cB from both sides to get pA cB >= pB cA. For positive times, this is pA/cA >= pB/cB: put the check with more rejection probability per minute first. Here A has 0.10/2 = 0.05 per minute and B has 0.50/5 = 0.10 per minute, so B goes first. This comparison assumes fixed check times and that stopping after a rejection is allowed.

HOW YOU COULD EXPLAIN IT ALOUD

I would compare the expected total time in each order. A first always costs two minutes, then ninety percent of items need another five, so that is six and a half minutes. B first always costs five minutes, then half need another two, giving six minutes. So I would choose B first. The result is not simply "run the faster check first": I am balancing the cost of a check against how often it avoids the remaining work.

Check the result. Every item takes at most 7 minutes. A-first times are 2 or 7; B-first times are 5 or 7. The calculated averages lie inside the appropriate ranges.

Avoid this mistake. Adding 2+5 assumes both checks always run. Multiplying both check times by pass probabilities forgets that the first check runs even on a failing item.

Use this idea elsewhere. For an early-stopping process, calculate the cost of each stage multiplied by the probability that the process reaches that stage.

QUESTION 08 / C / EXPECTED VALUE AND DECISIONS

Collecting all four symbols

Breaking an expected total into stages | Authored practice

The question

Each sealed packet contains one of four symbols, A, B, C or D. The symbols are equally likely, and packet contents are independent. You open packets until you have seen all four symbols. What is the expected number of packets opened, including duplicates?

THE ANSWER

The expectation is 25/3 packets, or about 8.33 packets.

How to work it out

1. Measure progress by the number of distinct symbols
The important state is how many different symbols have already appeared, not how many packets have been opened. Divide the process into four stages: get the first distinct symbol, then a second, then a third, then the last. The total packet count is the sum of the packet counts spent in those stages.

2. Work out the success chance in each stage
With no symbols yet, every packet gives a new symbol, so the first stage takes exactly one packet. With one distinct symbol collected, three of the four symbols are new, so the next packet progresses with probability 3/4. With two collected, the chance of a new one is 2/4 = 1/2. With three collected, only one of four symbols is new, so success chance is 1/4.

3. Derive the expected length of one stage
For a stage whose success chance is p on each independent packet, let E be the expected number of packets needed. We always open one packet. With probability 1-p it is a duplicate, leaving us in the same state with E packets still expected. So E = 1+(1-p)E. Rearranging gives pE=1 and E=1/p. This explains the reciprocal rather than asking you to memorise it.

4. Add the stage expectations
The four stage lengths average 1, 4/3, 2 and 4 packets. Their sum is 1+4/3+2+4 = 25/3. Adding expectations is valid even when random quantities are dependent; we do not need to prove independence between the stage lengths to add their averages. The final missing symbol accounts for four packets on average, which is why finishing the collection feels disproportionately slow.

QUESTION 08 / CONTINUED

Follow-ups and interview reasoning

Collecting all four symbols | Back to the question

Follow-up 1. You already have exactly two different symbols. How many additional packets do you expect to need?

Only the final two stages remain. Getting a third distinct symbol takes 1/(1/2) = 2 packets on average, and getting the fourth takes 1/(1/4) = 4. The expected additional number is 6. The number of duplicates opened earlier does not matter, because future packet contents are independent and the state is still "two types collected".

Follow-up 2. What is the probability you complete the collection in exactly four packets?

The first packet can be anything. The second must differ from the first, with probability 3/4. The third must be one of the two not yet seen, probability 2/4. The fourth must be the last remaining symbol, probability 1/4. Multiplying gives 1 x 3/4 x 2/4 x 1/4 = 3/32. Equivalently, the 4! = 24 orders containing each symbol once are favourable out of 4^4 = 256 equally likely packet sequences. Both give 24/256 = 3/32.

HOW YOU COULD EXPLAIN IT ALOUD

I would split the problem by how many distinct symbols I have. The first packet always progresses. The chance of getting a new type then falls to three quarters, one half and one quarter. For a fixed success chance p, the expected number of independent attempts is one over p, which I can derive by E equals one plus the failure chance times E. Adding the four stages gives one plus four thirds plus two plus four, or twenty-five thirds.

Check the result. At least four packets are needed, so an expectation below four would be impossible. An expectation above four is sensible because duplicates are possible. For just one symbol type, the same method gives one packet.

Avoid this mistake. Four symbols do not imply four packets on average. Also, 25/3 is an average over many runs, not a fractional number of packets that occurs in one run.

Use this idea elsewhere. Choose a state that captures the relevant progress. If each stage has a fixed success probability, derive its expected duration and add the stages.

QUESTION 09 / C / EXPECTED VALUE AND DECISIONS

Four people get their hats back

Indicator variables, linearity and overlapping events | Authored practice

The question

Four people each put one hat into a box. The four hats are then assigned uniformly at random, one to each person, with every one-to-one assignment equally likely. How many people do you expect to receive their own hat?

THE ANSWER

The expected number of correct hats is 1. This remains 1 for any positive number n of people.

How to work it out

1. Look at one person rather than the whole arrangement
Each person is equally likely to receive any of the four hats. Exactly one of those hats is their own, so that person has probability 1/4 of a correct match. It is easier to calculate this single-person probability than to list every possible number of matches.

2. Represent a match by zero or one
For person i, define Ii to be 1 if they get their own hat and 0 otherwise. Its expected value is 1 x 1/4 + 0 x 3/4 = 1/4. If X is the total number of correct hats, then X = I1+I2+I3+I4.

3. Add the contributions
E[X] = E[I1]+E[I2]+E[I3]+E[I4] = 4 x 1/4 = 1. This is called linearity of expectation: the expected sum is the sum of the expectations. We are not multiplying probabilities, so independence is not needed.

4. Explain why independence would be false
If the first three people all have their own hats, the fourth person must also have theirs. Their outcomes clearly constrain one another. Nevertheless, each individual still has a marginal probability 1/4 of being correct, and adding those expected contributions remains valid. For n people, each contributes 1/n, giving n x 1/n = 1.

QUESTION 09 / CONTINUED

Follow-ups and interview reasoning

Four people get their hats back | Back to the question

Follow-up 1. What is the probability that nobody gets their own hat in the four-person case?

There are 4! = 24 equally likely assignments. Count assignments with at least one correct hat using overlap correction. For each of four people, fixing their hat leaves 3! = 6 assignments: total 4 x 6 = 24, but assignments with multiple matches were counted repeatedly. Subtract the six possible pairs of fixed people, each leaving 2! = 2 assignments: subtract 12. Add back the four triples, each leaving one assignment: add 4. Finally subtract the one assignment fixing all four: subtract 1. Thus at least one match occurs in 24-12+4-1 = 15 assignments. No matches occur in 24-15 = 9, giving 9/24 = 3/8.

Follow-up 2. What does an expected value of one actually mean?

It means the number of correct hats averages to one over many independent assignments. It does not mean one person is correct every time. Zero, one, two or four people may be correct. Exactly three is impossible: if three hats are already correct, the remaining hat must also be correct. This is another useful check on any proposed distribution.

HOW YOU COULD EXPLAIN IT ALOUD

I would count the expected contribution of each person. A particular person has a one-in-four chance of receiving their own hat, so their expected contribution to the number of matches is one quarter. Adding four such contributions gives one. I do not need the matching events to be independent to add their expectations. If you ask for the probability of no matches, that is a different calculation; I would count permutations and correct for overlapping fixed-hat events.

Check the result. For two people, the only assignments have either two correct hats or none, each with probability 1/2. Their average is one, agreeing with the general result.

Avoid this mistake. Do not calculate P(no matches) as (3/4)^4. The events are dependent, even though linearity allowed us to ignore that dependence for the expected total.

Use this idea elsewhere. For an expected count, assign a zero-one indicator to each object and add individual probabilities. Save the full distribution for when it is actually needed.

QUESTION 10 / D / ALGEBRA AND RATES

Walking up a moving escalator

Relative motion and keeping track of elapsed time | Authored practice

The question

You walk up an escalator that moves upwards at a constant speed. You take 40 steps to reach the top. On a second trip, you move at twice your original stepping rate relative to the escalator and take 60 steps to reach the top. The escalator speed is unchanged. How many steps long is the escalator when stationary? Ignore entry and exit effects and assume each walking step advances one escalator-step length.

THE ANSWER

The escalator is 120 step lengths long. Its speed is twice your original stepping rate.

How to work it out

1. Separate your motion from the escalator motion
The distance from bottom to top comes from two contributions: the steps you walk relative to the moving surface, and the distance that the escalator carries you during the trip. Both contributions point upwards. The stationary length is the sum of those contributions, not just the number of steps you personally count.

2. Choose units that make the speeds manageable
Let your original stepping rate be v steps per second and the escalator speed be e step lengths per second. Let its length be N step lengths. In the first trip, taking 40 steps at rate v means the trip lasts 40/v seconds. During that time the escalator carries you e(40/v) step lengths. Thus N = 40 + 40e/v.

3. Write the second trip with the correct time
On the second trip your rate is 2v, and you personally take 60 steps. The trip therefore lasts 60/(2v) = 30/v seconds. The length equation is N = 60 + 30e/v. Although you took more steps, this trip took less time because you stepped twice as quickly.

4. Equate the two expressions for the same length
Both trips cover N, so 40+40e/v = 60+30e/v. Subtract 40 and 30e/v from the two sides to obtain 10e/v = 20. Hence e/v = 2. Substitute into either length equation: N = 40+40 x 2 = 120. The second gives 60+30 x 2 = 120 as a check.

QUESTION 10 / CONTINUED

Follow-ups and interview reasoning

Walking up a moving escalator | Back to the question

Follow-up 1. If your original walking rate is one step per second, how long does each trip take?

Now v=1 and e=2. The original trip lasts 40/v = 40 seconds. Your doubled-rate trip lasts 60/(2v) = 30 seconds. In the first trip you walk 40 step lengths and the escalator carries you 80; in the second you walk 60 and it carries you 60. Both totals are 120.

Follow-up 2. How long would the trip take if you stood still?

With v=1 in the preceding follow-up, the escalator speed is 2 step lengths per second. Standing still means the escalator supplies all 120 step lengths, so the time is 120/2 = 60 seconds. Without an absolute walking rate, we could only express this as 120/e = 60/v seconds. The original problem determines a speed ratio, not an absolute number of seconds.

HOW YOU COULD EXPLAIN IT ALOUD

I would add my own steps to the steps the escalator carries me through. Let my original rate be v and the escalator rate e. My first trip takes forty over v seconds, giving length forty plus forty e over v. The second takes sixty over two v, giving length sixty plus thirty e over v. Equating the lengths gives e over v equal to two, and the stationary length is 120. I would check that both trips produce the same total distance.

Check the result. Doubling your stepping rate should shorten the trip. It does: the time falls from 40/v to 30/v. The moving escalator also means your personally counted 40 or 60 steps are both less than its actual length of 120.

Avoid this mistake. The second trip does not last half as long merely because your stepping rate doubles. Your personal number of steps changes too; calculate time as steps divided by stepping rate.

Use this idea elsewhere. For motion on a moving surface, separate motion relative to the surface from motion of the surface, then give both the same elapsed time.

QUESTION 11 / D / ALGEBRA AND RATES

Two analysts sharing a backlog

Adding work rates and handling staggered starts | Authored practice

The question

Analyst A can clear a fixed backlog alone in 6 hours. Analyst B can clear the same backlog alone in 10 hours. Their rates are constant, they work on different items without duplication, and working together does not change either rate. How long will they take together?

THE ANSWER

Together they take 15/4 hours: 3 hours 45 minutes.

How to work it out

1. Turn completion times into rates
Call the whole backlog one job. A completes 1/6 of a job per hour, while B completes 1/10 of a job per hour. Rates describe how much each contributes during the same amount of time, so rates are the quantities that can be added.

2. Add the hourly contributions
Their combined rate is 1/6 + 1/10. Using denominator 30, this is 5/30 + 3/30 = 8/30 = 4/15 of the backlog per hour. The assumptions about no duplication and unchanged speeds are what make this addition valid.

3. Convert the combined rate back to a time
Time is work divided by rate. To complete one job at 4/15 jobs per hour takes 1/(4/15) = 15/4 hours. The fractional part is 3/4 of an hour, which is 45 minutes, giving 3 hours 45 minutes.

4. Check by counting actual completed work
In 15/4 hours, A completes (15/4)(1/6) = 5/8 of the backlog. B completes (15/4)(1/10) = 3/8. Their contributions sum to one full backlog, confirming that the completion time is consistent with both individual rates.

QUESTION 11 / CONTINUED

Follow-ups and interview reasoning

Two analysts sharing a backlog | Back to the question

Follow-up 1. A starts two hours before B joins. How long from A's start until the whole backlog is finished?

During the first two hours, A completes 2 x 1/6 = 1/3 of the job. The remaining work is 2/3. Once B joins, the rate is 4/15 per hour, so the remaining time is (2/3)/(4/15) = (2/3)(15/4) = 5/2 hours. Add the initial two hours: total time is 2+2.5 = 4.5 hours, or 4 hours 30 minutes. This is measured from A's start, not from B's arrival.

Follow-up 2. A third analyst C can finish the backlog alone in 15 hours. If all three start together, how long do they take?

C contributes 1/15 job per hour. The total rate is 1/6+1/10+1/15 = 5/30+3/30+2/30 = 10/30 = 1/3 job per hour. Completing one job therefore takes 3 hours. The result is smaller than the two-person completion time, as it should be when adding a productive analyst without reducing anyone else's rate.

HOW YOU COULD EXPLAIN IT ALOUD

I would measure the backlog as one unit of work. A does one sixth per hour and B does one tenth, so together they do four fifteenths per hour. Inverting gives fifteen quarters of an hour, which is three hours forty-five minutes. For the delayed start I would first subtract the one third that A finishes in two hours, then divide the remaining two thirds by the combined rate, and finally add the initial two hours.

Check the result. Together they must beat A's solo time of six hours. Because B is slower than A, two people together should take more than half of A's time, which would be three hours. The answer 3.75 lies between those bounds.

Avoid this mistake. Averaging 6 and 10 gives eight hours, which would make the team slower than A alone. Adding completion times or halving their average has no basis; add rates instead.

Use this idea elsewhere. For work, pumps, processing or motion, first express each contribution as an amount per unit time. Treat periods with different active contributors separately.

QUESTION 12 / D / ALGEBRA AND RATES

When do the clock hands meet?

Relative angular speed and exact time conversion | Authored practice

The question

At exactly 3:00, when will the minute hand next coincide with the hour hand? Assume both hands move continuously and at constant angular speeds.

THE ANSWER

They meet 180/11 minutes after 3:00, which is 16 minutes and 240/11 seconds later: approximately 3:16:21.82.

How to work it out

1. Measure the starting gap
At 3:00 the minute hand points at 12 and the hour hand points at 3. Taking clockwise as positive, the hour hand is 90 degrees ahead. The minute hand must make up that 90-degree lead while the hour hand continues moving.

2. Compute the two speeds in the same units
The minute hand travels 360 degrees in 60 minutes, or 6 degrees per minute. The hour hand travels 360 degrees in 12 hours, which is 720 minutes, so its speed is 360/720 = 1/2 degree per minute.

3. Use relative speed to close the gap
The minute hand gains on the hour hand at 6-1/2 = 11/2 degrees per minute. Therefore the catch-up time is distance divided by relative speed: 90/(11/2) = 180/11 minutes. Equivalently, at t minutes after 3:00 their angles satisfy 6t = 90+t/2, which yields the same equation.

4. Convert the fractional minute carefully
180/11 = 16+4/11 minutes. Multiply the fractional part by 60 seconds per minute: (4/11) x 60 = 240/11 seconds, approximately 21.82 seconds. This gives 3:16:21.82. The hands do not meet at 3:15 because the hour hand has moved forward during those fifteen minutes.

QUESTION 12 / CONTINUED

Follow-ups and interview reasoning

When do the clock hands meet? | Back to the question

Follow-up 1. How many coincidences occur from 00:00 inclusive to 12:00 exclusive?

After a coincidence, the minute hand must gain another full 360 degrees to coincide again. At relative speed 11/2 degrees per minute, the interval is 360/(11/2) = 720/11 minutes. In a 720-minute period, this produces eleven coincidences when the start is included and the endpoint is excluded: times 0, 720/11, ..., 10 x 720/11. The next one is exactly 720 minutes, which belongs to the following half-day.

Follow-up 2. What is the next time strictly after 3:00 at which the hands form a right angle?

At 3:00 the minute hand is 90 degrees behind the hour hand. That signed difference increases at 11/2 degrees per minute. To become 90 degrees ahead, it must gain 180 degrees, taking 180/(11/2) = 360/11 minutes. This is 32+8/11 minutes, or 32 minutes and 480/11 seconds, approximately 3:32:43.64. The word "strictly" excludes the right angle that already exists at 3:00.

HOW YOU COULD EXPLAIN IT ALOUD

At three o'clock the hour hand is ninety degrees ahead. The minute hand moves at six degrees per minute, but the hour hand also moves at half a degree per minute. So the gap closes at five and a half degrees per minute. Dividing ninety by five and a half gives 180 over 11 minutes, a little over sixteen minutes. I would convert the remaining four elevenths of a minute to seconds to get about 3:16:22.

Check the result. At 3:15 the minute hand is at 90 degrees, but the hour hand is at 97.5 degrees, so the meeting must be later. At t=180/11, both angles equal 1080/11 degrees, confirming the solution.

Avoid this mistake. Treating the hour hand as stationary gives 3:15. Also state the endpoint convention when counting events in a twelve-hour interval; including both midnight and noon counts the boundary twice across successive intervals.

Use this idea elsewhere. When one moving object catches another, work with the rate at which the gap changes. The same idea applies to angles, distances and repeated cycles.

QUESTION 13 / E / NUMBER STRUCTURE AND LOGIC

A number with several remainders

Combining divisibility conditions before searching | Authored practice

The question

Find the smallest positive integer that leaves remainder 1 when divided by each of 2, 3, 4 and 5, but is exactly divisible by 7.

THE ANSWER

The smallest such integer is 301. The only positive solutions below 1,000 are 301 and 721.

How to work it out

1. Translate the shared remainder
If a number N leaves remainder 1 on division by 2, then N-1 is divisible by 2. The same reasoning applies to 3, 4 and 5. Therefore N-1 must be a common multiple of all four numbers. Subtracting the common remainder turns four remainder conditions into one divisibility condition.

2. Find the smallest repeating block
The least common multiple of 2, 3, 4 and 5 is 60. To see why, a multiple of 4 already includes the factors required for 2. We still need a factor 3 and a factor 5, giving 4 x 3 x 5 = 60. Thus every candidate has the form N = 60k+1 for an integer k >= 0.

3. Search only the candidates that can possibly work
The positive candidates are 1, 61, 121, 181, 241, 301, and so on. Their remainders on division by 7 are respectively 1, 5, 2, 6, 3 and 0. Therefore 301 is the first candidate divisible by 7. Because the candidate sequence is increasing and includes every number satisfying the first four conditions, this also proves minimality.

4. Check the result directly
301 = 2 x 150 + 1; 301 = 3 x 100 + 1; 301 = 4 x 75 + 1; and 301 = 5 x 60 + 1. Also 301 = 7 x 43. All five requirements hold. A candidate that works is not enough on its own; the ordered candidate search is what establishes it is the smallest.

QUESTION 13 / CONTINUED

Follow-ups and interview reasoning

A number with several remainders | Back to the question

Follow-up 1. Find every solution below 1,000.

Any two solutions differ by a multiple of 60 because both are 1 more than a multiple of 60. They also differ by a multiple of 7 because both are divisible by 7. Since 60 and 7 have no common factor, their difference must be a multiple of 420. Adding 420 to a valid solution preserves every condition. Starting from 301 gives 301, 721 and then 1,141, so the only positive solutions below 1,000 are 301 and 721.

Follow-up 2. Can you avoid listing the first six candidates?

Yes. Reducing N=60k+1 modulo 7 gives 4k+1 divisible by 7, because 60 leaves remainder 4. We need 4k to leave remainder 6. Since 4 x 5 = 20 leaves remainder 6, k=5 works; successive solutions have k=5+7m. Substituting gives N=301+420m. This is shorter if modular arithmetic is comfortable, but the small candidate search is already a complete, transparent solution.

HOW YOU COULD EXPLAIN IT ALOUD

All four remainder-one conditions tell me that N minus one must be divisible by two, three, four and five. Their least common multiple is sixty, so I only need to inspect numbers sixty k plus one. The first few are one, sixty-one, one hundred twenty-one, one hundred eighty-one, two hundred forty-one and three hundred one. Three hundred one is forty-three times seven, so it works and is the smallest. I would then verify each remainder explicitly.

Check the result. The number must be odd because it leaves remainder one modulo two. It must also end in one or six from the condition modulo five, so being odd forces a final digit of one. Both 301 and 721 satisfy that quick check.

Avoid this mistake. Multiplying 2 x 3 x 4 x 5 gives 120, which is a common multiple but not the least one. Restricting candidates to 120k+1 could skip the smallest valid answer.

Use this idea elsewhere. When several divisibility conditions share a remainder, subtract that remainder first and combine the resulting multiples before doing any search.

QUESTION 14 / E / NUMBER STRUCTURE AND LOGIC

One heavy coin among nine

Designing a decision tree and proving a minimum | Authored practice

The question

Nine coins look identical. Exactly one is heavier than the others; the other eight all have the same weight. You have a balance scale with three possible results: left heavy, right heavy or balance. What is the minimum number of weighings that guarantees identifying the heavy coin, and what is the strategy?

THE ANSWER

Two weighings are necessary and sufficient. Split the coins into three groups of three, then split the remaining suspect group into three individual coins.

How to work it out

1. Use the possible outcomes of the tool
One weighing can return three outcomes. A useful first weighing should therefore divide the nine possibilities into three groups of similar size. If a branch leaves too many suspects, that branch determines the worst-case number of further weighings. We want to control the worst case, not only the fortunate balanced result.

2. Perform the first weighing
Label the groups A, B and C, each with three coins. Weigh A against B. If A is heavier, the heavy coin is in A. If B is heavier, it is in B. If they balance, all six weighed coins have the ordinary weight and the heavy coin is in C. Every possible result leaves exactly three suspects.

3. Perform the second weighing
Call the three remaining suspects x, y and z. Weigh x against y. If the left side is heavier, x is the heavy coin. If the right side is heavier, y is the heavy coin. If they balance, z must be the heavy coin. This identifies the answer in a total of two weighings in every branch.

4. Prove one weighing cannot be enough
There are nine possible identities for the heavy coin, and a single weighing supplies at most three distinguishable results. A strategy based on only one weighing therefore cannot assign a unique result to all nine possibilities. Two weighings can distinguish at most 3 x 3 = 9 outcome sequences, and the strategy above actually achieves that bound. The lower bound and the construction together prove the minimum is two.

QUESTION 14 / CONTINUED

Follow-ups and interview reasoning

One heavy coin among nine | Back to the question

Follow-up 1. What if there are 27 coins with one known to be heavier?

Three weighings are sufficient: weigh nine against nine, keeping the suspect group of nine; then three against three within that group, keeping three; then one against one. Three are also necessary because two weighings provide at most 3^2 = 9 possible result sequences, fewer than the 27 possible heavy-coin identities.

Follow-up 2. Why is weighing four coins against four an inferior first move for nine coins?

If the scale balances, the unweighed ninth coin is immediately identified. But if one side is heavier, four suspects remain. One more weighing has only three outcomes, which cannot always identify one heavy coin among four. The attractive one-weighing success in the balanced branch is paid for by a worse worst-case branch. Splitting 3,3,3 keeps every branch manageable.

HOW YOU COULD EXPLAIN IT ALOUD

The scale gives three outcomes, so I would split the nine coins into three equal groups. Weigh three against three: the heavy side contains the coin, or if they balance it is in the unweighed three. Then weigh one against one from the suspect group; a balance identifies the third coin. That uses two weighings. One cannot suffice because three possible results cannot distinguish nine possible heavy coins, so two is optimal.

Check the result. Check all three outcomes at both stages. The proof depends on knowing the unusual coin is heavier, and on all other coins having equal weight. Those assumptions make the heavy side identify the correct suspect group.

Avoid this mistake. Do not optimise only for a likely or convenient outcome. "Guarantee" asks for the maximum number of weighings over all possible locations of the coin.

Use this idea elsewhere. For identification problems, compare the number of possible states with the number of distinguishable outcomes, then build a balanced decision tree that achieves the bound.

QUESTION 15 / E / NUMBER STRUCTURE AND LOGIC

The lockers that remain open

Pairing divisors and using parity | Authored practice

The question

There are 100 closed lockers numbered 1 to 100. On pass 1, you toggle every locker. On pass 2, you toggle every multiple of 2. Continue: on pass k, toggle every locker whose number is a multiple of k, through pass 100. Toggling changes open to closed or closed to open. Which lockers are open at the end?

THE ANSWER

Exactly the perfect squares remain open: 1, 4, 9, 16, 25, 36, 49, 64, 81 and 100. There are ten.

How to work it out

1. Follow one locker instead of every pass
Take locker n. Pass k toggles it precisely when k divides n. For example, locker 12 is toggled on passes 1,2,3,4,6 and 12. So the number of times locker n is toggled equals the number of its positive divisors. This converts a long simulation into a property of one integer.

2. Reduce the final state to odd or even
A locker begins closed. One toggle opens it, two close it, three open it again, and so on. Therefore only the parity of the toggle count matters: odd means open, even means closed. We need to identify the integers with an odd number of divisors.

3. Pair divisors by multiplication
Whenever d divides n, the number n/d is another divisor. For 12, the pairs are (1,12), (2,6) and (3,4), giving an even total of six divisors. A pair can fail to contain two different numbers only when d=n/d, which means d^2=n. That happens exactly when n is a perfect square.

4. Explain why squares give an odd count
For n=36, divisor pairs are (1,36), (2,18), (3,12), (4,9) and the single middle divisor 6, paired with itself. All ordinary pairs contribute two divisors; the square root contributes one. Thus a square has an odd divisor count and ends open. A non-square has only ordinary pairs and ends closed. The squares up to 100 are 1^2 through 10^2, so there are ten open lockers.

QUESTION 15 / CONTINUED

Follow-ups and interview reasoning

The lockers that remain open | Back to the question

Follow-up 1. If the process uses 200 lockers and 200 passes, how many finish open?

The same divisor-pairing argument applies. We count positive squares no greater than 200. Since 14^2=196 and 15^2=225, the last open locker is 196 and there are fourteen open lockers. For a general positive integer N, the number is floor(sqrt(N)), meaning the greatest integer no larger than its square root.

Follow-up 2. In the 100-locker problem, what is open immediately after pass 99?

All lockers numbered at most 99 have already received every possible divisor pass, so only the nine squares 1,4,...,81 among them are open. Locker 100 has divisors 1,2,4,5,10,20,25,50 and 100. Before pass 100 it has received eight toggles and is closed. Thus exactly nine lockers are open after pass 99; pass 100 opens locker 100 and leaves the other lockers unchanged.

HOW YOU COULD EXPLAIN IT ALOUD

I would inspect one locker. Locker n is toggled once for each divisor of n, and it ends open only if that number of toggles is odd. Divisors usually pair as d and n divided by d. The only unpaired divisor occurs when those two values are equal, so n is a square. Therefore the open lockers are the squares from one squared through ten squared. That gives ten lockers, and I can verify the reasoning on a non-square like twelve and a square like thirty-six.

Check the result. Locker 1 is toggled once and ends open. A prime such as 7 is toggled on passes 1 and 7, twice, so it ends closed. Locker 4 is toggled on passes 1,2 and 4, three times, so it ends open.

Avoid this mistake. Do not count the square-root divisor twice. For a square, the pair (sqrt(n),sqrt(n)) refers to one divisor, which is exactly why the parity changes.

Use this idea elsewhere. When a process repeatedly flips a binary state, look for parity. When a count is almost always even, try pairing objects and identifying the cases where an object pairs with itself.

RESEARCH CONTEXT

Where this practice comes from

The underlying research is from 24 October 2026. This companion was written on 25 October 2026.

This is a new practice collection, built around the maths and reasoning skills identified in your original research. It does not add claims that these exact fifteen questions have been asked at Jane Street. Every prompt, follow-up and worked explanation in this companion is supplied for practice.

Jane Street / Trading Desk Operations Engineering

Official description of the initial problem-solving and critical-thinking interview, followed by a more technical interview. Supports the focus on modelling, trying methods and checking answers.

Jane Street / Probability & Markets guide

Official background on counting, conditional probability, expectation, recursion and interview communication. This guide is aimed at trading/research; it is a useful mathematical foundation rather than a TDOE question list.

Glassdoor / the original 33 TDOE reports

The seven pages already read in this chat informed the topic priorities. Candidate reports are anonymous recollections and can be incomplete or duplicated. The new questions here do not repeat those in the first PDF.

Glassdoor / TDOE intern questions

Additional role-related evidence for counting, probability and number constraints. Used to select relevant skills, not to assert that these newly written prompts are reported questions.

Wall Street Oasis / TDOE intern account

Corroborating candidate account describing first-round brainteasers and motivation questions before later coding. It remains a candidate recollection rather than an official specification.

A NOTE ON THE SPOKEN EXAMPLES

The interview paragraphs show how the reasoning can be communicated once you understand it. It is fine to pause, test a small case, revise a mistaken assumption or ask for clarification while solving. Practise explaining the reason for each step rather than reproducing the exact wording.